

# SOPHIA SIPAYBOUN

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## PROFESSIONAL SUMMARY

Incoming Bioinformatics graduate student and computer science graduate with hands-on experience building full-stack genomics tools, evaluating AI model outputs, and developing data pipelines. Proficient in Python, SQL, and data analysis, with demonstrated project experience in transcriptomic data analysis (RNA-seq, PCA/UMAP, differential expression) and a strong interest in data-intensive, interdisciplinary research.

## EDUCATION

**Northeastern University** | M.S. in Bioinformatics

Fall 2026 - May 2028

**St. Petersburg College** | B.S. in Software Engineering, GPA: 3.7/4.0

Aug 2022 – May 2026

## PROJECTS

**GeneScope — Transcriptomic Data Analysis Platform** | Python, Flask, Pandas, Plotly

2026

- Built a full-stack bioinformatics web application for transcriptomic data exploration, supporting gene expression dataset uploads, automated preprocessing, and quality control summaries.
- Implemented PCA and UMAP dimensionality reduction, heatmap generation, differential expression previews, and pathway enrichment workflows for RNA-seq datasets.
- Combined software engineering and computational biology to deliver an interactive genomics analysis platform capable of handling large-scale RNA-seq data.

**Data Collection System for The Leo Project** | Node-RED, MySQL

Jan 2024

- Built a real-time water distribution monitoring system supporting The Leo Project's work addressing water access challenges in Laikipia, Kenya.
- Simulated data flow with Node-RED for metrics such as water level and pH, and configured MySQL for real-time data storage.
- Developed a user-facing dashboard reflecting live database data, designed for future feature and analytics expansion.

## WORK EXPERIENCE

**Product Manager Extern** | BeReal

Jul 2026 – Sept 2026

Remote

- Designed and prototyped new app features supporting BeReal's growth objectives using industry-standard product tools.
- Facilitated focus groups and structured surveys to evaluate prototype usability, synthesizing participant feedback into actionable design recommendations.
- Delivered a professional proposal deck integrating user research, competitive benchmarking, usability testing outcomes, and a go-to-market rollout plan.

**Healthcare Data Analyst Extern** | TruBridge

Feb 2026 – May 2026

Remote

- Conducted exploratory data analysis on public healthcare datasets, examining infection trends across case counts, demographics, and time patterns.
- Used structured research and analytical tools to guide data discovery and support visualizations, dashboards, and reports on public health trends.
- Produced synthesis reports and stakeholder presentations covering infection trends, data limitations, population insights, and recommendations for action.

**Software Developer Intern** | Jetzy

Aug 2022 – May 2023

Remote

- Built an HTML-based visualization platform to analyze in-app user behavior.
- Used Python Flask to extract and process user data from a SQL server, supporting more data-driven decision-making.
- Developed a machine learning classification model to categorize online articles by topic, improving content recommendations.
- Built a web scraping pipeline to gather and classify articles by keyword, streamlining content categorization and boosting engagement.
- Conducted social media data mining to surface insights on organizational performance and activity.

## TECHNICAL SKILLS

**Programming:** Python, SQL, R, Java, C++, JavaScript, HTML, CSS

**Data & Machine Learning:** Data Analysis, Machine Learning, Model Evaluation, RLHF, Data Labeling, Classification Models

**Databases:** MySQL, MongoDB

**Frameworks & Tools:** Flask, Node.js, Spring, Flutter, Node-RED

**Other:** Data Pipelines, Web Scraping, Cloud Computing, Quality Assurance

## ORGANIZATIONS

**Stanford Data Ocean Bioinformatics Bootcamp (via Conectado)** | Member

Dec 2023 – Jan 2024

- Completed a career readiness program in collaboration with Stanford Data Ocean Bioinformatics Bootcamp.
- Earned a certificate in Fundamentals of Data Science in Precision Medicine and Cloud Computing.

**Rewriting the Code** | Member

May 2022 – Present

- Engaged in mentorship and networking opportunities with women in technology, strengthening professional development.
- Participated in technical workshops and career panels focused on coding, leadership, and industry best practices.